

SECTION THREE
Strength and
the Uphill Athlete



CHAPTER SIX

Strength Training for the Uphill Athlete

Athletes strength train to improve performance in their event and for injury prevention, not to become stronger in the gym. This is especially true for endurance athletes.

A lot of studies have tried to answer the question of how strong an endurance athlete needs to be. Two seemingly contradictory findings emerge from most of the studies:

- Endurance sports typically utilize up to only about 25 percent of an athlete's maximum single repetition strength, but do so for up to many thousands of repetitions.
- Nonetheless, increasing max strength does seem to have a positive effect on endurance.

While few studies of this type have been directed at mountain endurance athletes, it is safe to make the following assumption: Strength plays a big role in mountain sports where an athlete has to lift his or her body up and lower it down over potentially many thousands of meters.

While it makes intuitive sense that a certain minimal level of strength is required for endurance sports, if the second bullet point above were to be taken to its ultimate conclusion, Olympic weightlifters should be the fastest marathoners. Clearly this is not the case. Why?

Strength is largely a *neurologic* quality while endurance is largely a *metabolic* quality. Restating this: Strength depends on the brain's ability to recruit the greatest number of muscle fibers for a task. Endurance depends on the rate of metabolic

Previous page: Strength is a crucial contributor to an athlete's speed. Here a team races across the exposed ridge of Mont Coin, with Mont Blanc looming in the background, during the 2010 Pierra Menta skimo race. France. Photo: Jocelyn Chavy

Opposite: Forrest Coots earning his turns in the remote Cerro Castillo zone of Patagonia, Chile, while on a sixteen-day ski trip exploring the area. Photo: Jason Thompson



Katie French and Alyson Dimmitt Gnam put their strength to use dropping into a creek bed while circumnavigating Mount St. Helens, Washington. Photo: Steven Gnam

turnover of ATP molecules. We use the modifier *largely* above because these two qualities overlap and are interdependent in endurance sports.

Among similarly performing endurance athletes, the stronger ones tend to have better performance. This means that among all marathon runners with finishing times between say 2:10 and 2:15, you could list their running time and certain strength test results and see a strong correlation between performance and strength. Just being able to deadlift more does not guarantee that you will be a faster runner. But, if you can deadlift more than those with a similar running speed, chances are very good that you will also run faster than those people. More importantly, sport-specific strength correlates better than general strength when it comes to predicting performance. Studies have shown that sport-specific strength exercises like standing broad jump, the single-legged hop, and even the thirty-meter sprint are decent predictors of 5K running race times when comparing runners with similar times. These exercises are much more running specific than a general leg strength exercise like the deadlift.

So, how much strength is sufficient? The answer: No one really knows.

General Versus Specific Strength

Before going too deep into any discussion of strength training, we need to define some terms and concepts.

General strength is nonsport specific, meaning that it may bear little or even no similarity to the movement, speed, or range of motion made in the sport for which you are training. For an athlete, general strength provides a base for the more sport-specific strength training that follows and it makes the athlete less injury prone.

Specific strength is sport specific, meaning that it mimics quite closely (exactly, in some cases) the demands of the sport. This type of strength training has a direct bearing on the athlete's performance in that sport.

General Strength

Ski mountaineering, skimo, and mountain running are all locomotive sports involving single-legged propulsion. Effective movements in all of these rely on good hip stabilization in order for the prime-mover muscles to do their job best. So, even general strength training for these sports should place heavy emphasis on single-legged strength exercises.

Most of the popular gym-based strength exercises fall under the “general” category when training for sports.

So, why bother with general strength training at all? While there certainly are mountain athletes who avoid general strength training all together, you'd be wise to consider your own needs rather than blindly following what others do. If you are deficient in general strength, it could well benefit you to engage in a simple strength training program. It can make the sport-specific strength training (which relates directly to performance) more effective.

Keep the statement at the beginning of this chapter in mind as you develop a strength training program to improve performance in your sport. In the end, you don't care how much you bench-press or deadlift, right? You mainly care how fast you ski, run, or climb, right? Nonetheless it is easy to get swept up in the “gym strong” mentality and overdo the general strength. Spending too much time and effort on general strength can not only detract from more beneficial specific training time, but it is possible to become too strong with the bulky muscles that come with that excess strength.

Let's use one of the most popular general strength movements, the back squat, as an example of what we are talking about. The back squat is a popular general strength movement for many sports because it engages some of the biggest prime-mover muscles in the hips and legs as well as the core. Improving this strength up to a point can translate to better sports performance. But where is that point? Again, no one really knows.

In strength, speed, and power sports like sprinting, the field events of throwing and jumping, or American football, increasing squat strength will correlate quite closely with improved performance. For endurance sports where we propel



Diego Pazos winning the Mont Blanc 80K in Chamonix in 2015. France. Photo: Martina Valmassoi

ourselves with one leg at a time, an exercise like the squat will fit only into the general category.

Even though we can't make categorical recommendations regarding how much general strength you need, we can make some suggestions. If you are a mountain endurance athlete who can't do a half squat with your own body weight carried on your shoulders, you could probably gain performance and durability with added strength. But it is doubtful that being able to squat twice your body weight will add anything to your performance and may in fact detract from it. Your performance would no doubt benefit more from spending the additional hours doing specific strength exercises, rather than trying to increase your squat strength by twenty pounds.

Similarly, the pull-up is a very good upper-body general strength exercise, and if you are a skimo racer who can't do a single pull-up, you would probably see gains by improving this general strength.

However, no athlete, especially the amateur with job, school, and family obligations, has unlimited time or energy to devote to training. You need to examine the strength demands of your sport before diving into a bench press program when your goal is improving your VK time.

Specific Strength

These workouts should model the ranges of motion and speeds of your event quite closely. Why? The brain's muscle-recruitment patterns are a learned skill. Doing very heavy and slow movement in a semispecific strength exercise like a single box step is a great way to build general strength in those muscles, but at some point in your progression you need to transfer that box-step strength into a specific exercise like Hill Sprints. The speed of muscle contraction is much faster, and those motor neurons need to be trained to produce the sport-specific movement pattern and contraction speed, and not just slow strength. This is the justification offered by endurance athletes who predominately use specific strength methods.

A simple exercise that meets the needs of the mountain sports being covered in this book is sprinting or bounding uphill. Both the speed and gradient add the resistance and result in the desired strength-training effect using the sport-specific movement patterns for all these sports under consideration.

Economy and Muscular Endurance

While exceptional levels of strength are not required (and may even be detrimental) for athletes participating in an endurance sport, *sufficient* and especially,



Anne Gilbert Chase training at The Mountain Project in Bozeman, Montana.
Photo: Jason Thompson

sufficient specific strength do contribute to better movement economy. How? Increasing strength allows for longer strides with less effort. It takes a smaller percentage of an athlete's maximum strength to propel him- or herself farther in one stride. Especially for ultrarunners, the decrease of stride length during races is one of the most powerful predictors of race results. Muscular endurance training can greatly improve the ability to ward off the sort of late race neuro-muscular fatigue brought on by pounding down hills. We will dig deeper into muscular endurance training later.

Speed and Strength

Speed of movement is directly related to strength in that movement. In general, the stronger you are, the easier it will be for you to overcome the resistance to your movement. For mountain sports, overcoming gravity when going uphill, or resisting it when going down, requires strength. So, gaining strength, especially specific strength, up to a point benefits you. But, too much can hurt if it adds mass and lowers the strength-to-weight ratio. The quality we call muscular endurance is more important in successful training for all endurance events than pure strength. As we will show, there are many ways to develop it—from specific muscular endurance, circuit-style workouts, to uphill Zone 3 interval training, all the way to long runs on hilly terrain.

Muscular Endurance and Strength

The boundary between strength and endurance becomes blurred when we talk about a concept called muscular endurance (also called local muscular endurance and strength endurance). As the name implies, muscular endurance relates to the amount of strength that can be maintained over an event-specific length of time. Because of event specificity, this term takes on very different meanings for a 100K runner versus the VK runner.

The shorter the event, the larger the role played by muscular strength; the longer the event, the greater importance of muscular endurance. But the governing concept in both cases is the same. The reason for the merging of strength with endurance is that to complete many repetitions (be that 100 reps at 50 percent of your maximum strength or 20,000 reps at 20 percent) depends both on the maximum strength and also the endurance (metabolic) qualities of the most powerful muscle fibers needed to complete the task. In chapter eight we will look closer at ways to assess and improve muscular endurance.



Verónica Bravo running through the fog above Palma de Mallorca, Spain.
Photo: Martina Valmassoi

Summary

Some important points to keep in mind as you read ahead are:

- Athletes, and especially endurance athletes, use strength training as one of several tools. They do not get stronger just to be stronger. They get stronger to enhance their performance.
- Not all strength training is accomplished in the gym by lifting weights.
- Gaining strength without adding to overall body mass is critically important for endurance sports. After all, you have to carry those muscles around.
- Improving muscular endurance can have a very strong positive effect on improving performance and should be considered an important part of most training programs. By enhancing fatigue resistance it maintains higher levels of running economy in long events.

Next page: Clare Gallagher chasing her why during a winter training run, up to Chasm Lake, Colorado. Photo: Fredrik Marmsater

ATHLETE STORIES

Don't Forget to Pack the Why

Clare Gallagher

Aside from the myriad preparations I do for an ultra-race, from the physical to the mental, I also bring my *why* to the start line. I'm never going to run a 50K, let alone a 100-miler, just for the hell of it. If anyone aloofly says that they run these just for kicks, they're lying.

We all need a *why*. Mine has evolved over the few years I've been in the sport. I ran my first 50-miler—in Thailand—only six months after my last collegiate track race, and then I tackled my first 100-miler less than two years after that. My *why* began as a mixed curiosity of, *Can I do this?* Plus, *I really want to run in the Golden Triangle*. Eventually it got to, *Why not try a 100-miler?* combined with, *Well, I have nothing else going for me right now*.

But the most common *why* to the many ultras I've run, aside from the fact I'm a little sick in the head, is that I really love running long across mountainous terrain knowing that it will never get any easier. Sure, I can be more prepared, fitter, experienced, and tapered before an ultra, but I've yet to finish a race and think, *Wow, what a delightful cruise that was!* It's always going to suck somehow. And I love it.

You'll sprain an ankle. Your hydration flask will explode and you'll lose all of your Coke on a devastatingly hot trail section. You'll chafe the skin off your inner thighs. You'll come to in a bathtub outside Auburn, California, at 3:00 a.m. while your mom bathes your raw pink body, and you're screaming because *something* is wrong with your leg and there's hardly any skin left in your butt crack. You'll feel tired and slow and fat and stupid and want to quit. You'll think the world is ending. You'll contemplate finding broken glass just so you can step on it to make the misery end once and for all.

You'll pee blood. You'll poop blood. You'll crap your pants and dig so many holes that you'll lose count. While flying downhill, you'll hit a sharp rock exactly on the blister on your big toe, causing it to sear so badly that you'll see black dots. You'll yell at your selfless

friend who gave up her weekend to pace you. Then you'll cry, begging her for forgiveness. You'll threaten your boyfriend: *If you don't find me some ibuprofen...* You'll sob at evergreens, appalled that they are not helping you through this. You'll laugh with aspens, complimenting their breathtaking golden leaves.

You'll wonder if you've lost your mind.

You'll eat a jar of frosting and finish it with a liter of Coke. You'll fantasize about vegetable broth and salt pills. Your headlamp battery will die in the first mile of your first 100-miler. You'll fall face-first on a flat fire road, smashing your already-tweaked patella. You'll eat a pack of seaweed-flavored potato chips in your first ultra, cementing your addiction to the sport. You'll vomit. You'll DNF a race—heartbroken and physically broken.

Then, approximately thirty minutes to three weeks after the last agony-filled shitstorm, you find yourself sheepishly researching your next race. How are we so stupid?

Maybe within all of the sucking, all of the suffering, there's magic that we tap into during ultras.

I keep coming back for the moments of utter serenity, deep in a forest or high on an alpine trail, or even just grinding away on a godforsaken fire road. I live for the quietness. Where it's just me, a little human creature moving by her own steam somewhere on this beautiful planet. I patter along far enough that maybe, using a powerful telescope, an alien from another world could see the route I traveled across Earth that day. The alien doesn't have to know about the anguish ... that's mine to cherish. ■

Clare Gallagher is a trail runner and a lunatic. She thought running track and cross-country at Princeton wasn't cool enough, so she moved to Thailand to teach English and run her first ultra. Subsequently, she gained ten pounds of fried rice and Thai sweets (not a joke). She's been significantly less injured since then (also not a joke). So please eat some ice cream.



